

MKM-ST-001 — Threshold Sensitivity Analysis

Component: Stress Test Pipeline — MKM-ST-001 **Scope:** How the pipeline's flood decision responds to error in the per-gauge **severe-threshold calibration** — the single most material uncertain input to the FloodPoly classifier surrogate that drives every flood label in the stress catalogue. **Date:** 2026-07-30. **Method:** closed-form, data-free; composes the calibrated FloodPoly coefficients (`config.models.flood_poly_coeffs`) via `models.stress.sensitivity`.

1. Why the severe threshold is the material input

The pipeline labels a flood through FloodPoly, a logistic surrogate for the per-gauge GBM classifier:

$$P(\text{flood}) = \text{logistic}(a*h + b*t + c*h*t + d*h^2 + e*t^2 + f)$$
$$h = \ln(w / s) \quad t = \ln((\text{hour} + 1) / T)$$

Both features are gauge-independent, and the water level w enters **only** through the log-margin $h = \ln(w/s)$ against the severe threshold s . That threshold is a per-gauge calibration number, unvalidated on synthetic data, so a relative error in s (or, symmetrically, in the stress water level w) shifts h and moves the flood label. This analysis holds the coefficients fixed and perturbs s about a gauge at $w = s = 6.58$ m at the final storm hour.

2. The transition band is sharply sensitive

At the severe threshold the surrogate reads $P(\text{flood}) = 0.292$ (conservative by design — the offset f keeps it below one-half at $w = s$). A plus/minus 5% error in the threshold moves it enormously:

percentile	severe s (m)	log-margin h	$P(\text{flood})$	move vs median
p05	6.25 (-5%)	+0.051	0.631	+0.34
p50	6.58	0.000	0.292	—
p95	6.91 (+5%)	-0.049	0.087	-0.21

A 5% calibration error swings the flood probability across a **0.09–0.63** range — tens of percentage points — from one gauge's threshold alone. The local slope dP/dh peaks near six around $w = s$: a one-percent error in threshold or water level is worth roughly six-tenths of a percentage point of flood probability at its steepest.

3. The tails are robust

Away from the threshold the logistic saturates and the response collapses. At $w/s = 1.35$ (water 35% over threshold) $P(\text{flood}) = 0.997$ and the slope falls to 0.05 — a 5% calibration error is then worth a fraction of a percentage point. Well-separated events — clearly safe or clearly flooded — are effectively immune to threshold error; the model's own card records the same shape, over-predicting in the alert-to-severe transition and under-predicting at the tails.

4. Implication for the catalogue

Because the entire stress catalogue is labelled through this one surrogate, its fragility is concentrated exactly where events cluster near the severe threshold — the alert-to-severe band the pipeline exists to probe. A systematic mis-calibration of s across gauges would bias the **count** of stressed floods, not merely individual probabilities, and would do so hardest for the marginal events that matter most.

5. What this does and does not establish

The probabilities and slopes are exact readings of the calibrated surrogate, not sampled estimates. They bound the *consequence* of a threshold error; they do not size the error. On synthetic gauges the severe threshold is an assumed input, so its realism is unvalidated (validation questions VQ-005 data quality and VQ-002 accuracy — outstanding). The control implication is direct: threshold calibration is the highest-value validation target for the pipeline, and it matters most for gauges whose stressed water levels sit near their severe marks.

Reproduce: `models.stress.sensitivity – threshold_sensitivity, flood_probability_distribution`; figures from the calibrated FloodPoly coefficients of 2026-07-30.